and developed simplex method (1947) to solve complex real time applications. Linear programming is one of the best optimization technique from theory, application and computation point of view.

Definition:

Linear Programming Problem(LPP) is a mathematical technique which is used to optimize (maximize or minimize) the objective function with the limited resources.

Mathematically, the general linear programming problem (LPP) may be stated as follows.

Maximize or Minimize

$Z = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n$

Subject to the conditions (constraints)

$a_{11} x_1 + a_{12} x_2 + \ldots + a_{1n} x_n \leq (or = or \geq) b_1$

$a_{21} x_1 + a_{22} x_2 + \ldots + a_{2n} x_n \leq (or = or \geq) b_2$

$\ldots \quad \ldots. \quad \ldots. \quad \ldots. \quad \ldots$

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$a_{m1} x1 + a_{m2} x_2 + \ldots + a_{mn} x_n (\leq or = or \geq) b_m$

$x_1, x_2 \ldots\ldots x_n \geq 0$

Short form of LPP

Maximize or Minimize $Z = \sum_{j=1}^{n} c_j x_j$

Subject to $\sum_{j=1}^{n} a_{ij} x_j \leq (or = or \geq) b_i$,

i = 1,2,3,…,m ... (1)

and $x_j \geq 0$ ... (2)

Some useful definitions:

Objective function:

A function $Z = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n$ which is to be optimized (maximized or minimized) is called objective function.

Decision variable:

The decision variables are the variables, which has to be determined $x_j, j = 1,2,3,\ldots,n$, to optimize the objective function.

Constraints:

There are certain limitations on the use of limited resources called constraints.

$\sum_{j=1}^{n} a_{ij} x_j \leq (or = or \geq) b_i, \; i = 1,2,3,\ldots, \; m$ are the constraints.

Solution:

A set of values of decision variables x_j, j=1,2,3,…, n satisfying all the constraints of the problem is called a solution to that problem.

Feasible solution:

A set of values of the decision variables that satisfies all the constraints of the problem and non-negativity restrictions is called a feasible solution of the problem.

Optimal solution:

Any feasible solution which maximizes or minimizes the objective function is called an optimal solution.

Feasible region:

The common region determined by all the constraints including non-negative constraints $x_j \geq 0$ of a linear programming problem is called the feasible region (or solution region) for the problem.



Linear Programming Problem helps a farmer to produce the best crop by minimising risk and maximizing profit.

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